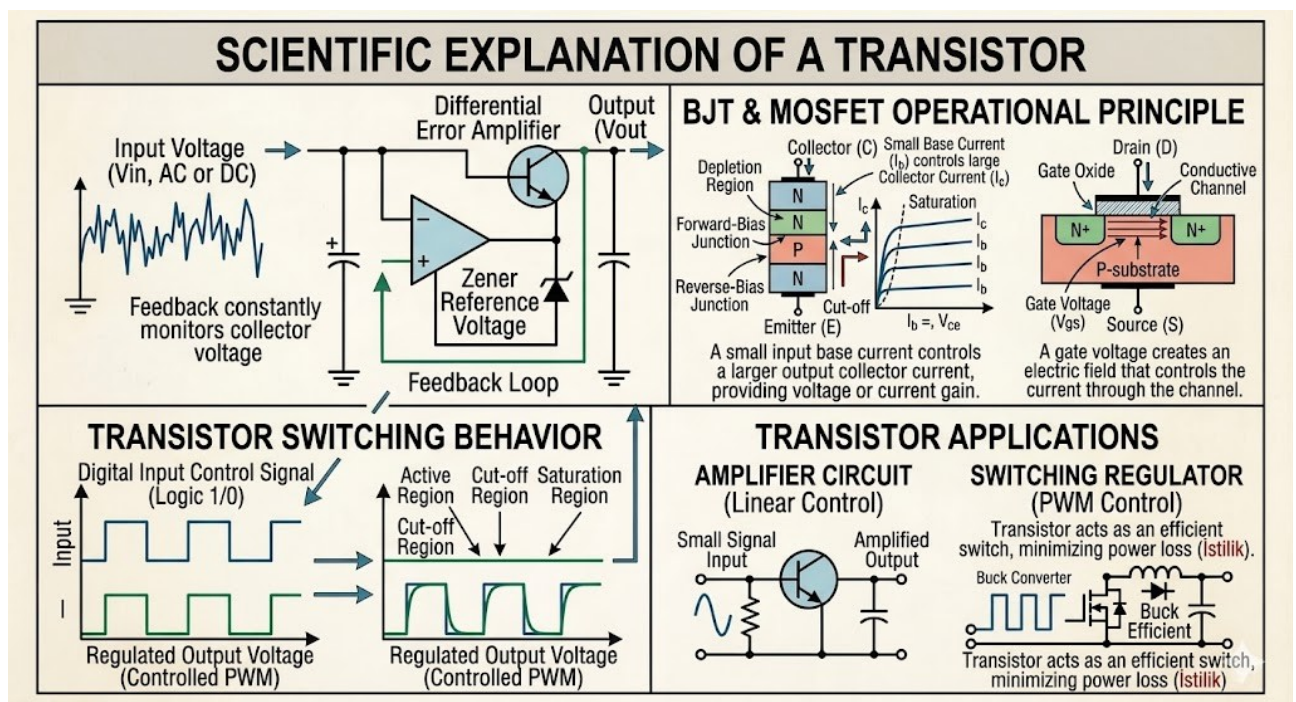


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A **transistor** is an active three-terminal semiconductor device used to amplify or switch electrical signals and power. It acts as the fundamental building block of modern digital and analog electronic circuits.

- **Input Voltage (V_{in}) & Control Stage:** The circuit handles an input signal which requires modulation or regulation. Capacitors are placed in parallel at the input and output rails to decouple high-frequency electrical noise and smooth out voltage ripples.
- **The Transistor Pass Element:** Positioned directly between the input and output, the transistor acts as an electronically controlled valve. By altering the electrical charge on its control terminal, it regulates the heavy flow of main current passing through it.

- **Feedback Loop Monitoring:** A differential error amplifier monitors the output voltage (V_{out}) via a feedback loop, comparing it against a highly stable **Zener Reference Voltage**.
 - **Dynamic Control:** If the output voltage fluctuates due to a changing load, the error amplifier instantly shifts the control current/voltage sent to the transistor. This forces the transistor to adapt its internal resistance, ensuring a flat, perfectly regulated output.
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III SECTION 2: BJT & MOSFET OPERATIONAL PRINCIPLES

Transistors generally fall into two primary categories, differing completely in how they are controlled:

1. Bipolar Junction Transistor (BJT) — Current Controlled

- **Structure:** Composed of three alternating semiconductor layers: **Collector (C)**, **Base (B)**, and **Emitter (E)** arranged in an NPN or PNP sandwich configuration.
- **Physics:** When a tiny forward-biasing current is injected into the thin Base terminal (I_b), it lowers the depletion region barrier between the emitter and collector. This allows a much larger collector current (I_c) to flow. The device operates as a **current-controlled current source**, providing high current and voltage gain.

2. Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) — Voltage Controlled

- **Structure:** Composed of three main terminals: **Drain (D)**, **Gate (G)**, and **Source (S)**, where the Gate is completely isolated from the conductive channel by a microscopic glass-like layer of Gate Oxide (SiO_2).
 - **Physics:** Because of the insulated gate, a MOSFET draws zero continuous input current. Instead, applying a Gate Voltage (V_{gs}) generates a powerful electric field across the oxide layer. This field attracts charge carriers to form an inversion layer (**Conductive Channel**) beneath the substrate, allowing current to flow effortlessly between Drain and Source. It acts as a **voltage-controlled current source**.
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✂ SECTION 3: TRANSISTOR SWITCHING & APPLICATIONS

Depending on how it is driven, a transistor operates across three distinct operational regions: **Cut-off** (fully off), **Active** (linear amplification), and **Saturation** (fully on).

1. Amplifier Circuits (Linear Control / Active Region)

- **Operation:** The transistor is biased strictly within its Active region. A minute continuous input signal (like an audio waveform) is fed to the control terminal.
- **Behavior:** The transistor linearly mirrors this small wave into a massive, high-power replica at the output, making it ideal for signal processing and audio amplification.

2. Switching Regulators (PWM Control / Cut-Off & Saturation)

- **Operation:** Instead of operating smoothly in the middle, a digital Pulse Width Modulation (PWM) control signal rapidly forces the transistor to jump between two extremes: fully **ON** (Saturation) and fully **OFF** (Cut-off).
- **Behavior:** When fully OFF, current is zero, meaning power dissipation is zero. When fully ON, its internal resistance is near-zero, meaning the voltage drop across it is negligible, which minimizes power loss as heat ("İstilik"). This ultra-fast switching mechanism forms the backbone of high-efficiency Buck and Boost power converters.

6. BJT VS. MOSFET: CRITICAL COMPARISON (Mühəndislik Müqayisəsi)

When designing custom PCB circuits in simulation tools like Proteus or Tinkercad, selecting the right transistor type is essential:

Feature	Bipolar Junction Transistor (BJT)	MOSFET
Control Signal	Current-driven: Requires a continuous base current (I_b) to stay ON.	Voltage-driven: Requires only a voltage potential (V_{gs}) at the gate; draws zero continuous current.
Switching Speed	Slower (due to charge storage in the base layer).	Extremely Fast: Can switch millions of times per second (MHz range), ideal for PWM.

Feature	Bipolar Junction Transistor (BJT)	MOSFET
Power Loss	Higher power loss due to continuous base current and higher saturation voltage.	Very Low Power Loss: It has a very tiny static resistance ($R_{ds(on)}$) when fully open.
Thermal Runaway	Vulnerable: As it gets hotter, it conducts more current, which can lead to self-destruction.	Stable: Naturally resists current increase when temperature rises, making it safer for high-power loads.

7. KEY SPECIFICATIONS ON A DATASHEET (Məlumat Vərəqi Parametrləri)

When choosing a specific transistor part number (like the **2N2222 NPN BJT** or **IRF540N N-Channel MOSFET**), engineers look for these safe limits:

- **V_{CEO} / V_{DSS} (Maximum Breakdown Voltage):** The absolute maximum voltage that can be applied across the main terminals (Collector-Emitter or Drain-Source) when the transistor is turned OFF. Exceeding this will permanently destroy the semiconductor crystal.